AWS Networking Practical

Implementation of VPC, Network ACLs and Security Groups for Public-Private EC2 Communication

AWS Practical Documentation November 3, 2026

1 Aim

To create a VPC with public and private subnets, configure a custom Network ACL and Security Groups, deploy two EC2 instances, and verify private-network communication between them.

2 Introduction

Think of this setup as two roles inside one private network: a public-side EC2 host that starts the test, and a private EC2 host that has no public IP address. The exercise builds that layout, then checks whether the first host can reach the second over its private address. The design keeps the private instance off public IPv4 addressing while allowing a specifically scoped connection to be tested.

The VPC and subnet counts came from AWS's *VPC and more* wizard. A custom Network ACL was also created, but the screenshots do not reveal its rule entries or subnet links. Those gaps are identified in the report rather than filled with assumptions.

3 Prerequisites

- An AWS account with access to the VPC and EC2 consoles.
- An SSH client and access to the public EC2 instance for running the connectivity test.

4 AWS Services and Concepts Used

Amazon VPC: The private network boundary for this lab. It holds the subnets and gives the two instances a shared network environment.

Subnet: A smaller IP range inside the VPC, placed in an Availability Zone. The public and private subnets give the two instances different network roles.

Network ACL: A traffic-control layer at the subnet boundary. It evaluates inbound and outbound traffic; the rule details and subnet links for this lab's custom ACL are not visible in the evidence.

Security Group: A virtual firewall attached to an EC2 network interface. It controls traffic for the associated instance, rather than applying to an entire subnet.

EC2: AWS's virtual server service. Here, one instance starts the test and the other provides the private destination.

5 Implementation

5.1 VPC creation with VPC and more

Instead of building the network one resource at a time, the VPC wizard applies the requested layout as a single setup flow. The selected plan used one Availability Zone, one public subnet, one private subnet, and no NAT Gateway.

Table 1: VPC wizard configuration

Setting	Value
VPC creation option	VPC and more
VPC name shown after creation	NACL-SG-Lab-vpc
VPC ID	`vpc-0b991cb380b365156`
VPC IPv4 CIDR	`10.0.0.0/16`
IPv6	Disabled
Availability Zones	1
Public subnets	1
Private subnets	1
NAT Gateway	None

The wizard's name-tag input is `NACL-SG-Lab`. The created VPC appears in the console as `NACL-SG-Lab-vpc`. The public and private subnets provide separate placement for the two instances. No NAT Gateway was selected, and no NAT configuration is claimed in this report.

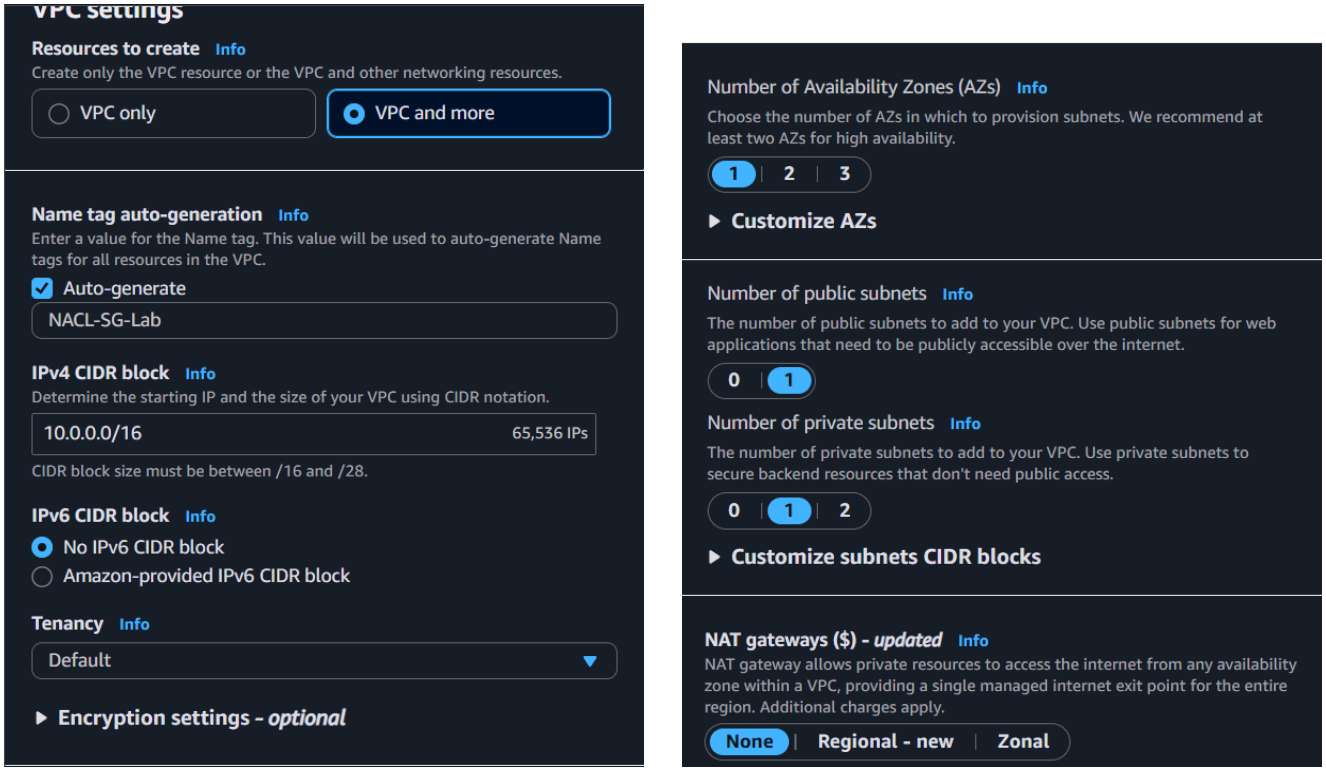


Figure 1: VPC and more wizard selections: VPC settings, subnet counts, Availability Zones, and NAT Gateway choice.



Figure 2: Created VPC listed as available, with its name and VPC ID.

5.2 Custom Network ACL creation

After the VPC was in place, a custom Network ACL named `Salil-Ke-Liye` was created for `NACL-SG-Lab-vpc`. A Network ACL filters traffic at the subnet boundary, with inbound and outbound rule sets. The evidence confirms the ACL name and VPC, but does not show its rules or subnet associations, so no rule numbers, protocols, ports, or allow/deny actions are assumed.

Table 2: Visible custom Network ACL details

Setting	Value
Name	`Salil-Ke-Liye`
VPC	`NACL-SG-Lab-vpc`
Inbound rules	Not available in the provided evidence
Outbound rules	Not available in the provided evidence
Subnet associations	Not available in the provided evidence

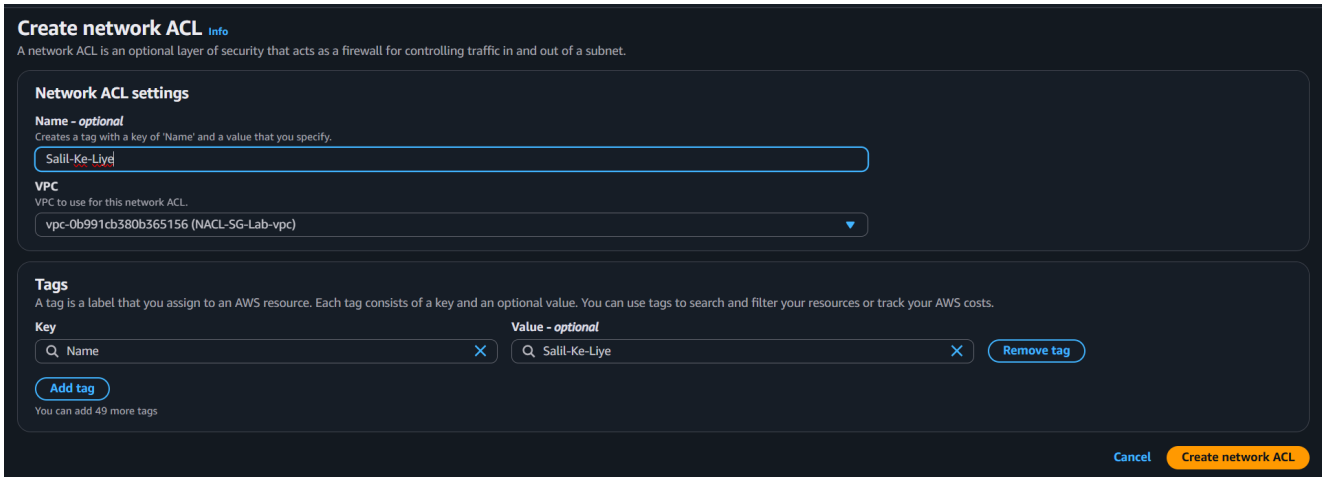


Figure 3: Custom Network ACL form showing its name and selected VPC. The image does not show ACL rules or subnet associations.

5.3 Security Group creation and configuration

The distinction is easiest to see by where each control sits: a Network ACL guards a subnet boundary, while a Security Group is attached to an instance's network interface. Security Groups are stateful; Network ACLs use inbound and outbound rule sets at the subnet level. The screenshots show the Security Groups' inbound entries only, so the tables below do not infer any outbound rules.

Table 3: Public Security Group inbound rule

Security Group	Type / protocol	Port	Source
`NACL-SG-Lab public` ID: `sg-01e0b08812205bc3d`	SSH / TCP	22	`125.22.234.212/32` (My IP)

For the public instance, SSH is limited to the displayed `/32` source address. The private instance has two inbound allowances: SSH and all ICMP from the public Security Group. Using that group as the source identifies traffic from network interfaces associated with it; it does not open the rule to an arbitrary

IP range. The reference makes the intended public-to-private path explicit, but successful delivery still depends on routing, applicable Network ACL rules, and the destination host's settings.

Table 4: Private Security Group inbound rules

Security Group	Type / protocol	Port range	Source
`NACL-SG-Lab Private` ID: `sg-07f2e5780989c6485`	SSH / TCP	22	Public Security Group `sg-01e0b08812205bc3d`
`NACL-SG-Lab Private`	Custom ICMP – IPv4 / all ICMP	All	Public Security Group `sg-01e0b08812205bc3d`

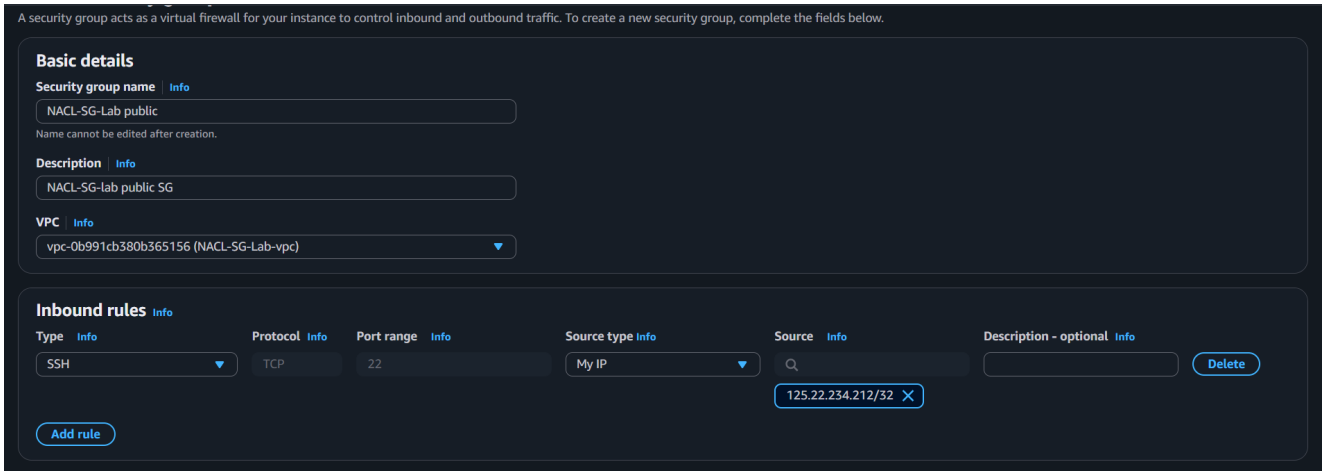


Figure 4: Public Security Group creation form with the SSH rule and My IP source.

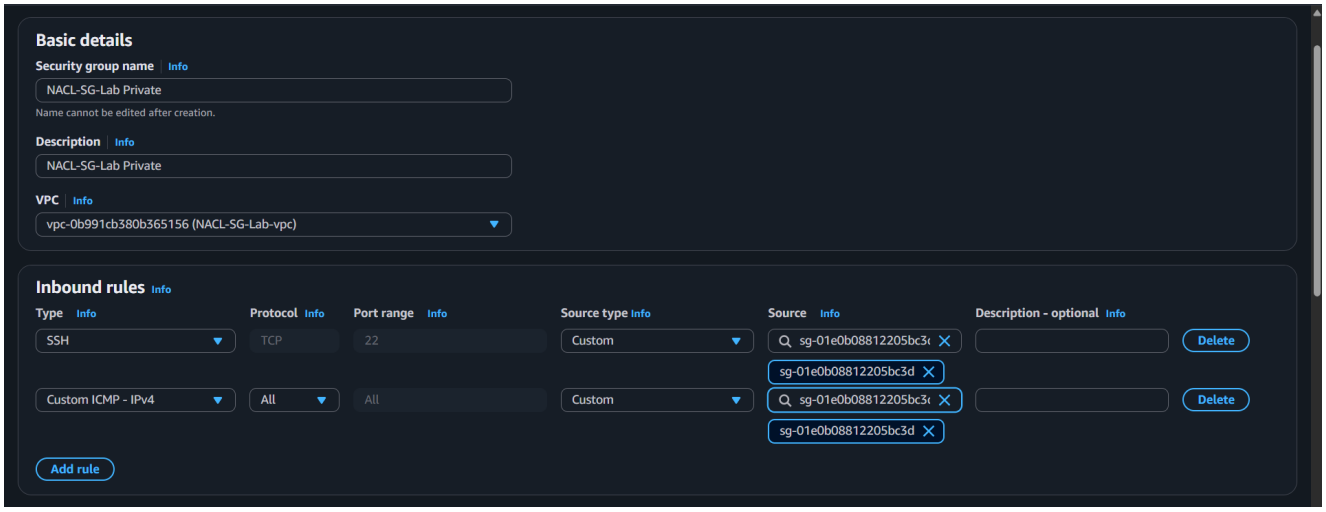


Figure 5: Private Security Group creation form with SSH and all-ICMP IPv4 rules sourced from the public Security Group.

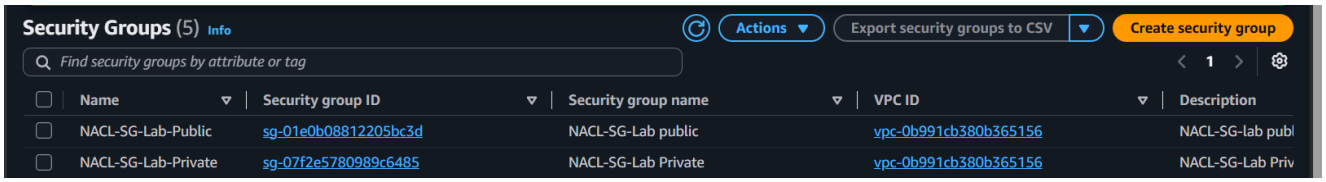


Figure 6: Security Group list showing both groups, their IDs, and their association with the lab VPC.

5.4 EC2 instance deployment

The network now has two clear roles: the public EC2 instance is the test source, and the private EC2 instance is the destination. Both are `t3.micro` instances in Availability Zone `ap-south-1a`. Public-IP assignment is enabled for the public instance and absent for the private one. This gives the public host a public IPv4 address when the surrounding network permits it, while the private host is not directly addressed through public IPv4. Since both sit in the same VPC, they can use private addresses to communicate when routing and security rules allow.

Table 5: EC2 instances launched for the practical

Instance	Type	Subnet	AZ	Public IP
NACL-SG-Lab Public EC2	`t3.micro`	Public	`ap-south-1a`	Enabled
NACL-SG-Lab Private EC2	`t3.micro`	Private	`ap-south-1a`	Not assigned

The public-instance network settings screenshot shows the lab VPC, public subnet, enabled public-IP assignment, and selected public Security Group. The instance list shows both instances running, their instance types and Availability Zone; the private instance has no public IPv4 DNS value shown.

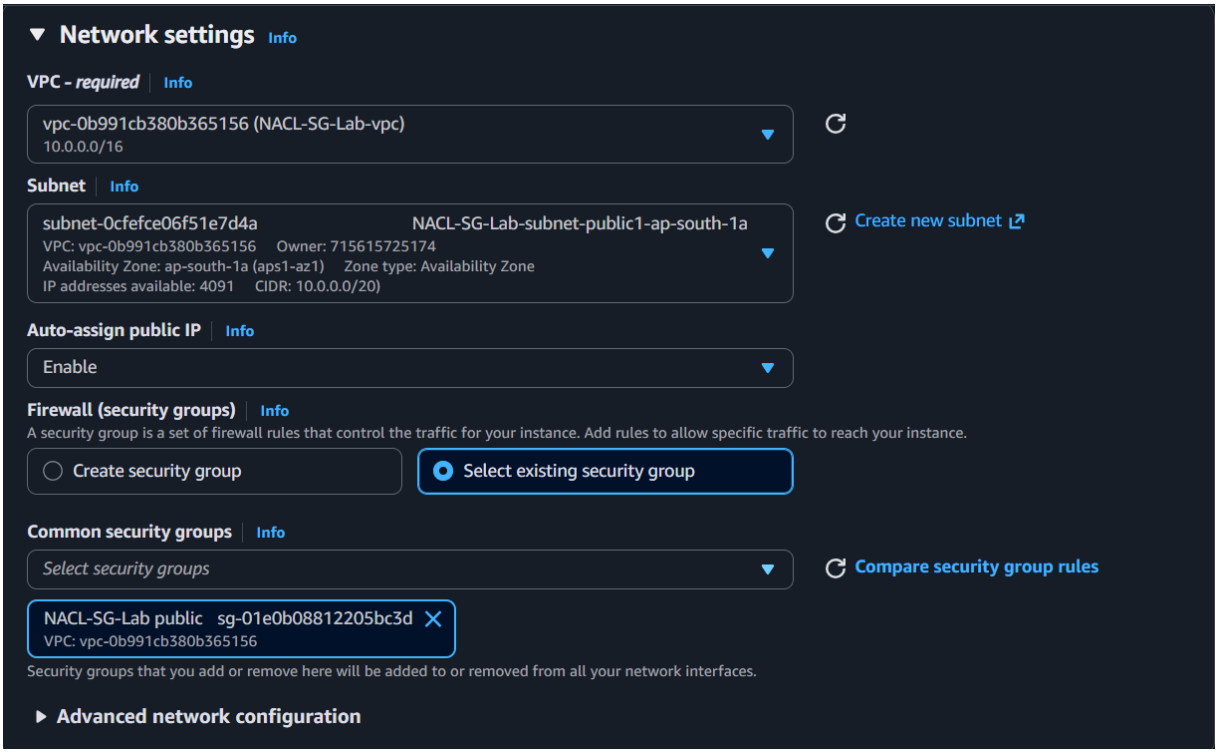


Figure 7: Public EC2 network settings: lab VPC, public subnet, public-IP assignment enabled, and public Security Group selected.

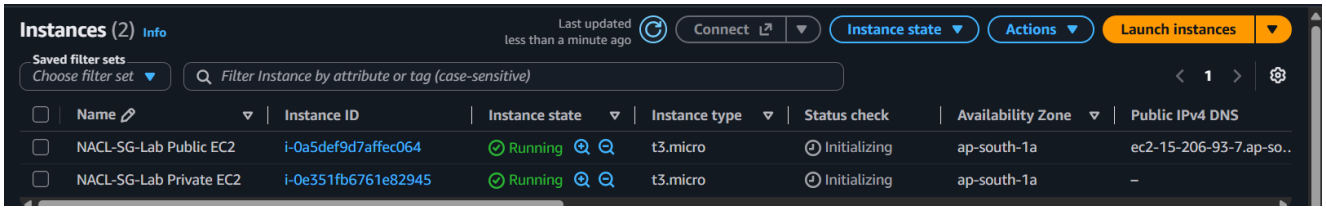


Figure 8: EC2 console list showing the public and private instances running in `ap-south-1a`.

5.5 Connectivity testing

From the public EC2 instance, the private instance's private IP address was tested with `ping`:

> **Test path** Public EC2 ⟶ private instance at `10.0.138.19`; ICMP echo requests are used to check this path.

The captured terminal output reports:

```
8 packets transmitted, 8 received, 0% packet loss
```

All eight echo requests received replies, with no packet loss in the captured run. That is evidence of ICMP connectivity from the public instance to `10.0.138.19` over the private network path. It is a useful end-to-end check, but it does not independently verify every Security Group or Network ACL rule, and it says nothing about internet connectivity.

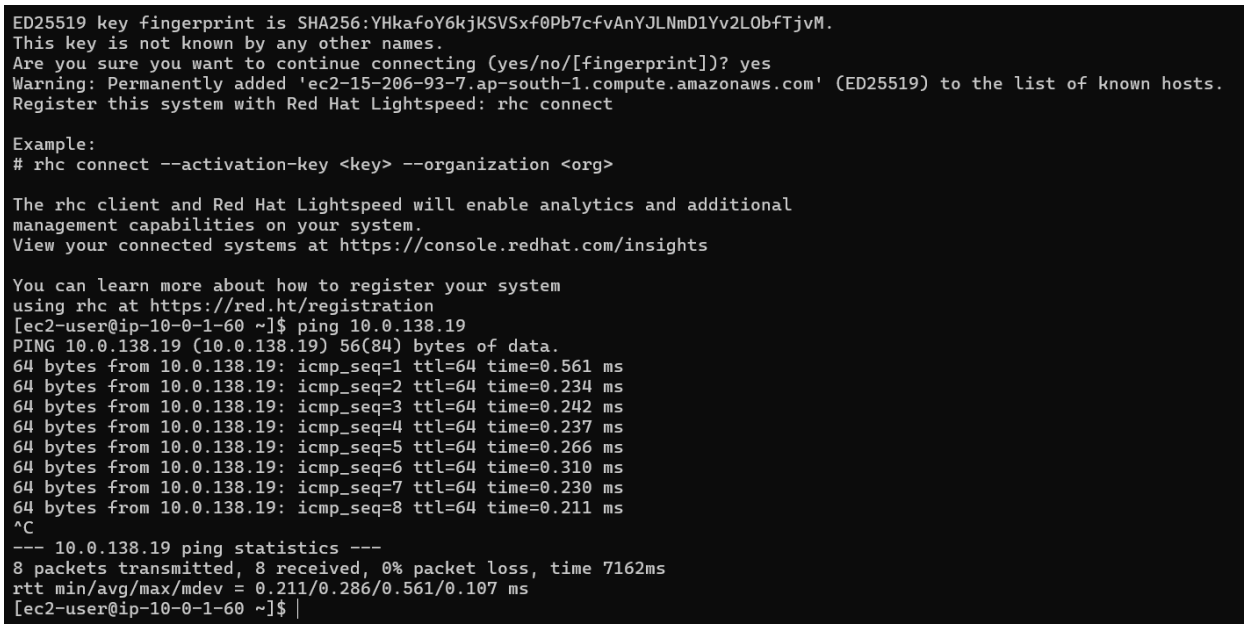


Figure 9: Terminal output from the public EC2 instance showing eight successful ping replies from the private IP address.

6 Result

The configured VPC contains the documented public and private subnet layout, and the two EC2 instances were launched in their respective roles. The custom Network ACL is recorded with the visible name and VPC. Most importantly, the captured ping run returned eight replies from the private IP with no packet loss, demonstrating successful private-network ICMP communication for this test.

7 Conclusion

The main takeaway is that a private instance can remain without a public IP and still be reachable from a public-side instance over the VPC's private network, provided the relevant path and security

controls permit it. The Security Group source reference demonstrates a focused way to describe that allowed source. The Network ACL's rule behavior and subnet associations remain unverified because those screens were not included in the evidence.

8 Glossary

VPC Virtual Private Cloud; a logically isolated virtual network in AWS.

CIDR Classless Inter-Domain Routing notation used to express an IP address range, such as `10.0.0.0/16`.

Subnet A subdivision of a VPC's IP address range associated with an Availability Zone.

Network ACL A subnet-level control with inbound and outbound traffic rules.

Security Group An instance/network-interface-level virtual firewall that controls allowed traffic.

Inbound rule A rule describing traffic allowed to enter the protected resource or network boundary.

Outbound rule A rule describing traffic allowed to leave the protected resource or network boundary.

Public IP An IPv4 address that can be used to address an instance from outside its VPC when network routing and security permit.

Private IP An address used for communication within private network ranges, including communication between resources in a VPC when routing and security permit.

ICMP Internet Control Message Protocol; the protocol used by the ping test to send echo requests and receive replies.

Availability Zone An isolated location within an AWS Region in which AWS resources, including subnets, can be placed.